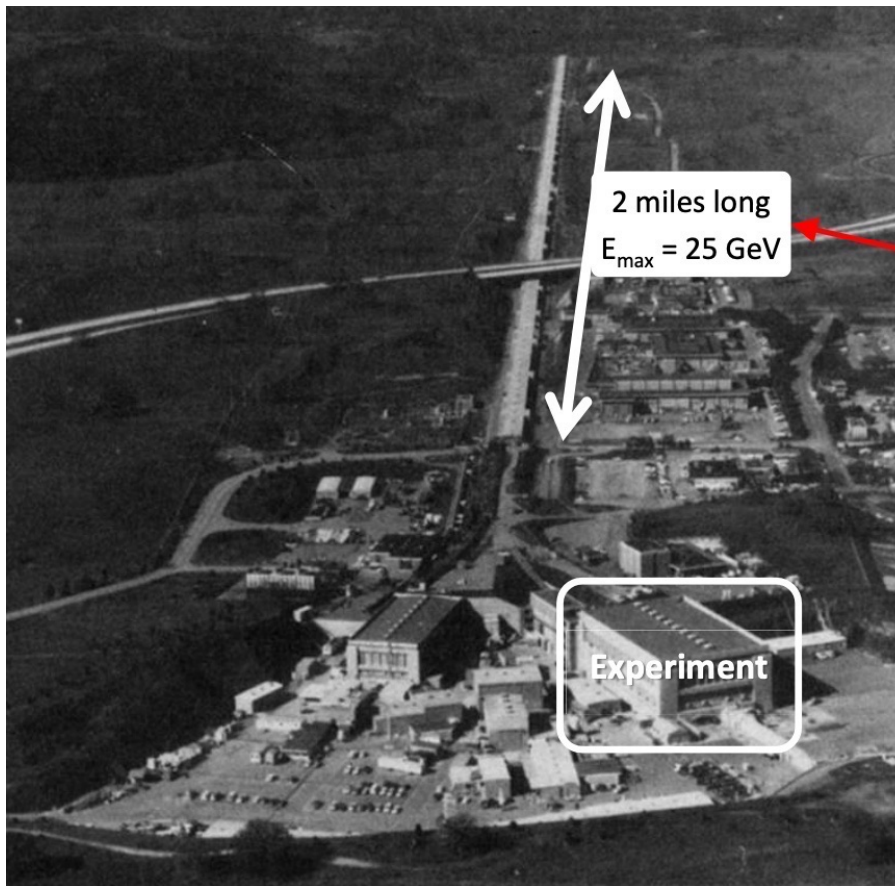


Deep Inelastic Scattering

$$e^- + p \rightarrow e^- + X$$

with increasing energy E of e^-
 $\Rightarrow$ proton not pointlike
 already at $E \approx 0.5$ GeV

Linear Accelerator @ SLAC $E \rightarrow 25$ GeV



$e^- + p \rightarrow e^- + X$
 look at X
 measure
 E, E', θ

spectrometers
 to see outgoing
 electron.

$\Rightarrow$ determine
 Q^2, ν

3 spectrometers:

$$Q^2 = 4EE' \sin^2 \frac{\theta}{2}$$

1) 1.6 GeV $\theta \approx 34^\circ$

8 GeV } $\theta > 12^\circ$
 20 GeV }

SLAC: Stanford Linear Accelerator
 Center

electric current $I = \frac{Q}{\Delta t} = \frac{dq}{dt}$

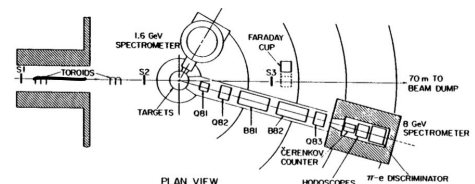
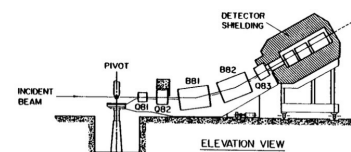
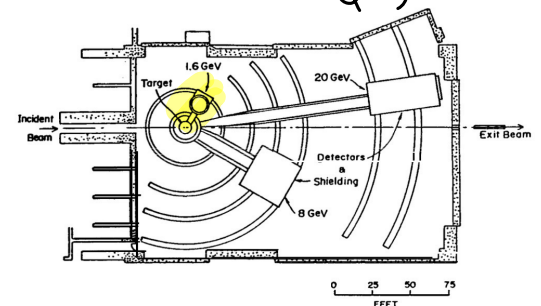


Fig. 2. (a) Plan view of End Station A and the two principal magnetic spectrometers employed for analysis of scattered electrons. (b) Configuration of the 8 GeV spectrometer, employed at scattering angles greater than 12° .



$$e^- + p \longrightarrow e^- + p \quad \text{elastic}$$

$$e^- + p \longrightarrow e^- + X \quad \text{inelastic}$$

$$e^- \xrightarrow{E} \cdot \nearrow_{\theta} e^- \quad E'$$

$$N_{\text{inelastic}} = \sigma_{\text{inelastic}} \cdot \mathcal{L}_{\text{inst}} \cdot \Delta t$$

↳ instant. luminosity.

counting # events in spectrometers.

→ need $\mathcal{L}_{\text{inst}}$ of the accelerator.

$$N_{\text{elastic}} = \sigma_{\text{elastic}} \cdot \mathcal{L}_{\text{inst}} \cdot \Delta t$$

well known
and calculated
from theory

σ_{Mott}

$$\text{Prob. (elastic)} = \frac{\sigma_{\text{elastic}}}{\sigma_{\text{elastic}} + \sigma_{\text{inelastic}}}$$

N_{elastic} : measured

Elastic events:

$$Q^2 = 2M\nu$$

$$= 2M(E - E')$$

$\Rightarrow$ measure L_{int} of the accelerator from elastic events.

$$\Rightarrow \sigma_{inelastic} = \frac{N_{inelastic}}{L_{int} \Delta t}$$

$$Q^2 = 4EE' \sin^2 \frac{\theta}{2}$$

$$W^2 = M^2 - Q^2 + 2ME'$$

$$\frac{d^2\sigma}{d\Omega dE'} = \frac{\# \text{ events}}{(dt \sin \theta) (E' \text{ bin})}$$

1) Elastic scattering decreases with E

2) @ fixed W value

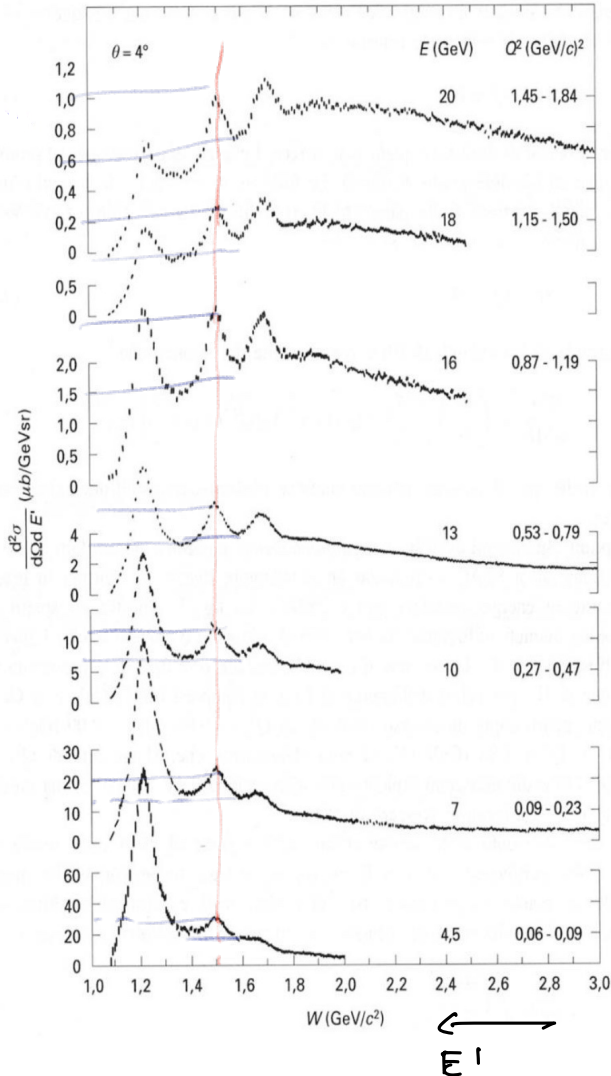
$\sigma \searrow$ with $E \nearrow$

3) Fixed E, θ

σ smaller for heavier W

4) very large W

$$\frac{d\sigma}{d\Omega dE'} \sim 1-2 \mu\text{b}/(\text{GeV} \cdot \text{sr})$$



Expected Cross Section

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} \sim \frac{\alpha^2}{4} E'^2 \cos^2 \frac{\theta}{2} \frac{E'}{E}$$

point like

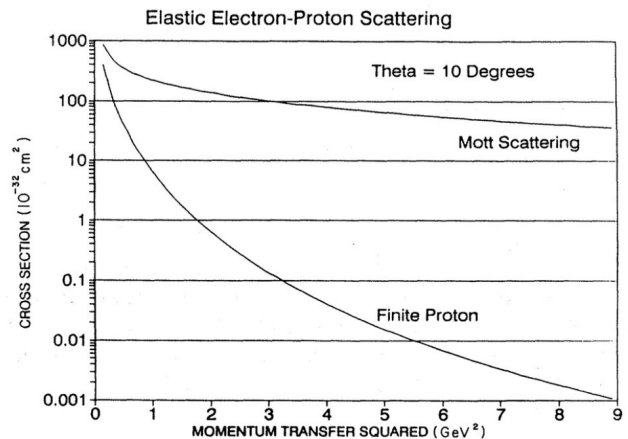


FIG. 4. Elastic scattering cross sections for electrons from a "point" proton and for the actual proton. The differences are attributable to the finite size of the proton.

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{structure}} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} \left[\frac{G_E^2 + \tau G_M^2}{1 + \tau} + 2\tau G_M^2 \tan^2 \frac{\theta}{2} \right]$$

$$\tau = \frac{Q^2}{4M^2}$$

G_E : electric form factor

G_M : magnetic form factor.

$$\frac{Q^2}{M^2} \ll 1 \quad (Q^2 \rightarrow 0) : \quad \text{small } Q^2 \text{ regime}$$

$$\tau \rightarrow 0$$

$$G_E(Q^2) \approx 1 - \frac{1}{6} Q^2 \langle r^2 \rangle$$

(discovery of proton structure)

$$\frac{Q^2}{M^2} \gg 1, \quad \tau \gg 1$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{structure}} \approx \left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} 2\tau G_M^2 \tan^2 \frac{\theta}{2}$$

$$G_M \approx \frac{g}{1 + (Q^2/0.71)^2}$$

$g = 2.79$ (anom. magn. mom of proton)

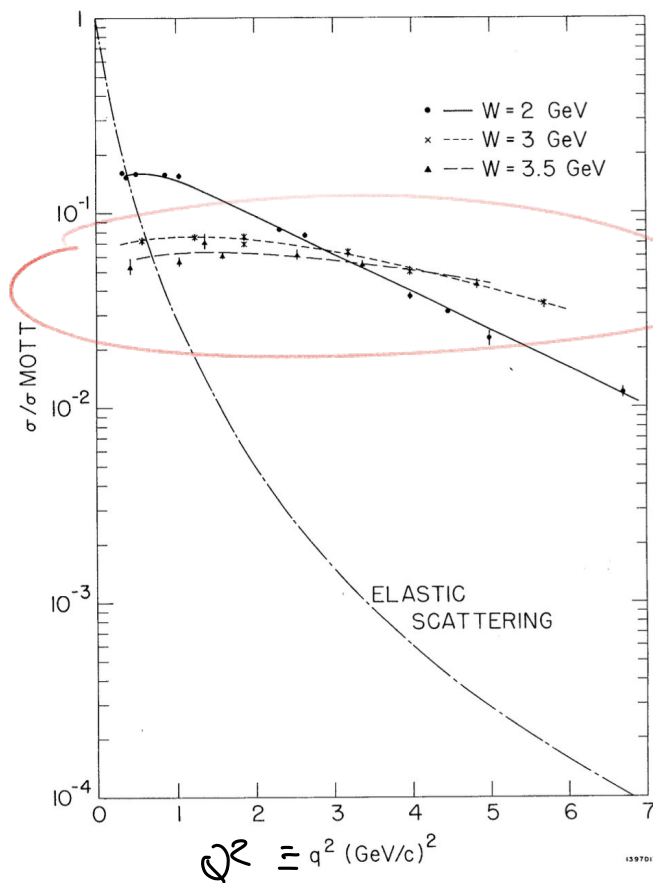
(dipole approx)

$$Q^2 \gg M^2 \quad G_M^2 \approx \frac{1}{Q^8}$$

$$\left. \frac{d\sigma}{d\Omega} \right|_{Q^2 \gg M^2} \approx \left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}} \cdot \frac{1}{Q^8}$$

$\Rightarrow$ experimentally measure $\frac{d\sigma}{d\Omega}$

$$\Rightarrow \frac{\left. \frac{d\sigma}{d\Omega} \right|_{\text{meas}}}{\left. \frac{d\sigma}{d\Omega} \right|_{\text{Mott}}}$$



almost constant
for energy q^2

$$\frac{\frac{d\sigma}{d\Omega}|_{\text{elastic}}}{\frac{d\sigma}{d\Omega}|_{\text{mott}}}$$

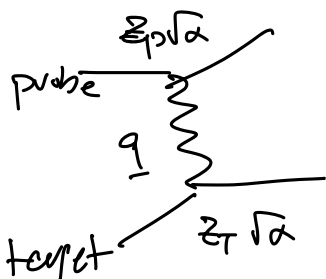
Elastic scattering on
proton with structure
and magnetic moment

$$\frac{d\sigma}{d\Omega} \Big|_{\text{point like}} = \overbrace{\frac{\alpha^2}{q^4} E'^2 \cos^2 \frac{\theta}{2} \frac{E'}{E}}^{\text{Mott}} |F(Q^2)|^2$$

$$\frac{\frac{d\sigma}{d\Omega}|_{\text{meas}}}{\frac{d\sigma}{d\Omega}|_{\text{mott}}} \approx \text{const} \Rightarrow (F(Q^2))^2 \approx 1$$

point like target

$$\frac{\frac{d\sigma}{d\Omega}|_{\text{meas}}}{\frac{d\sigma}{d\Omega}|_{\text{mott}}} \approx 10^{-1} \neq 1$$

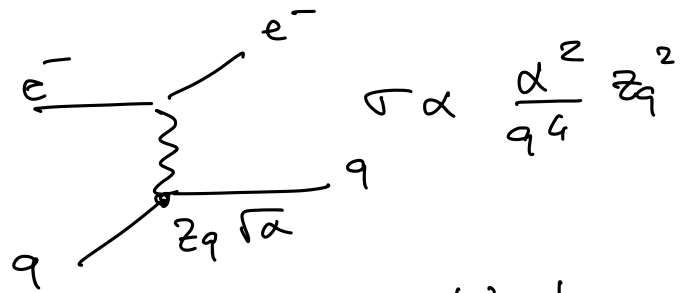


$$\sigma \propto \frac{(Z_P Z_T \alpha)^2}{q^4}$$

$$e^- : Z_P = 1 \quad \text{proton} : Z_T = 1$$

$$\sigma_{ep} \propto \frac{\alpha^2}{q^4}$$

$$\sigma_{ep} / \sigma_{\text{Mott}} < 1 \Rightarrow$$



pointlike object
 $S=1/2$ $Q_e = Z_q e$

$$Z_q < 1, \quad m \ll m_{\text{proton}}$$

$\Rightarrow$ proton made of partons/quarks

$$\left| \frac{e^-}{q} \right|^2 / \left| \frac{e^-}{e} \right|^2 \propto Z_q^2 < 1$$

$\Rightarrow$ experimental proof of partons/quarks.

$$\left| \frac{e^-}{P} \right|^2 = \left| \frac{e^-}{\text{parton}} \right|^2 + \left| \frac{e^-}{\text{meson}} \right|^2 + \dots$$

$\Downarrow$
 Deep Inelastic $e^- + p$ scattering $\equiv$ incoherent sum of elastic scattering with partons.

$\Rightarrow$ Quark Parton Model

$$\left. \frac{d^2\sigma}{dQ^2 dE'} \right|_{\text{DIS}} = \frac{\alpha^2}{Q^4} E'^2 \left[W_2(Q^2, \nu) \cos^2 \frac{\theta}{2} + 2W_1(Q^2, \nu) \sin^2 \frac{\theta}{2} \right]$$

structure functions

$$\left. \frac{d^2\sigma}{dQ^2 dE'} \right|_{\text{elastic}} = \frac{\alpha^2}{Q^4} Z_q^2 E'^2 \left[\cos^2 \frac{\theta}{2} + \frac{2Q^2}{4m^2} \sin^2 \frac{\theta}{2} \right] \delta\left(\nu - \frac{Q^2}{2m}\right)$$

m : mass of parton/quark.

$$\nu - \frac{Q^2}{2m} = 0 \Rightarrow 2m\nu = Q^2 \Rightarrow \text{elastic } e-q \text{ scattering.}$$

$$\frac{Q^2}{2m\nu} = 1$$

$$m = X\mu$$

μ : mass of target/proton.

$$\Rightarrow \frac{Q^2}{2X\mu\nu} = 1 \Rightarrow X = \frac{Q^2}{2\mu\nu} \quad \text{Bjorken variable.}$$

$$W_1(Q^2, \nu) \Big|_{x_0} = \frac{Q^2}{4m^2} \delta\left(\nu - \frac{Q^2}{2m}\right) = \frac{Q^2}{4\mu^2 x^2} \delta\left(\nu - \frac{Q^2}{2\mu x}\right)$$

$$x \in [0, 1]$$

$$x_0 = \frac{Q^2}{2\mu\nu}$$

No knowledge of distribution of x for pions.

$f(x)$: pion density function.

$f(x)dx \equiv \text{prob of having a pion with}$
 $x \in [x_0, x_0 + dx]$

$$\int_0^1 f(x) dx = 1 \quad \left[\text{required as prob. density function} \right]$$

$$W_1(Q^2, \nu) = \int_0^1 dx f(x) \frac{Q^2}{4\mu^2 x^2} z_q^2 \delta\left(\nu - \frac{Q^2}{2\mu x}\right)$$

$$\int A(x) \delta(g(x)) dx = \frac{A(x_0)}{|g'(x_0)|}$$

$$g(x_0) = 0$$

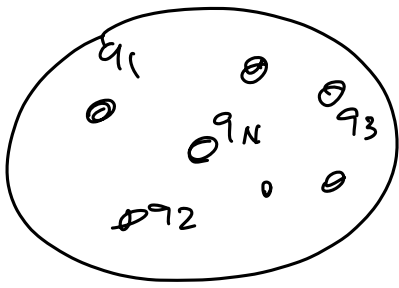
$$g(x) = \nu - \frac{Q^2}{2\mu x}$$

$$A(x) = \frac{f(x)}{x^2}$$

(do the derivatives)

$$W_1(Q^2, \nu) = z_q^2 \frac{f(x)}{2\mu}$$

for one pion.



$f_i(x)$: for each type i of partons.

$$W_1(Q^2, \nu) = \sum_i \sum_{\text{partons}} z q_i^2 \frac{f_i(x)}{2M}$$

$$W_2(Q^2, \nu) = \sum_i \sum_{\text{partons}} z q_i^2 \frac{x}{\nu} f_i(x)$$

$$F_1(x) \equiv M W_1(Q^2, \nu) = \frac{1}{2} \sum_i \sum_{\text{partons}} z q_i^2 f_i(x)$$

$$F_2(x) \equiv \nu W_2(Q^2, \nu) = \sum_i \sum_{\text{partons}} z q_i^2 x f_i(x) = x \sum_i \sum_{\text{partons}} z q_i^2 f_i(x)$$

$$F_2(x) = 2x F_1(x) \quad \text{Callan-Gross Relation.}$$

F_2 does not depend on Q^2
In principle

F_1, F_2 depend on Q^2, ν

$$F(x, y) = \frac{x^2 + y^2}{xy}$$

